

Primitive-Ring Symmetry Index Specification

Stage 7R: Compact Catalog-Bound Derived Ring Action

mdstats

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1 Purpose and boundary

`primitive_ring_symmetry.py` stores the exact action of one `PeriodicNetSymmetry` on one `PrimitiveRingCatalog`. It is a derived index, not a field of the core net-symmetry result.

Runtime/API target:

mdstats 0.19.19a0

The ownership chain is

```
PeriodicNetView
  -> PeriodicNetSymmetry
PrimitiveRingCatalog
  -> PrimitiveRingIndex
      + PeriodicNetSymmetry
      -> PrimitiveRingSymmetryIndex
```

This split prevents a net automorphism group from depending on a particular ring search bound and binds every ring orbit explicitly to the exact catalog that supplied it.

2 Motivation

A periodic-net automorphism group is defined by the decorated graph view. A primitive-ring action additionally depends on:

- which ring family was enumerated;
- the minimum and maximum search sizes;
- resource-completeness metadata; and
- the exact ring-catalog digest.

Persisting ring data inside `PeriodicNetSymmetry` merged two scientific sources, repeated complete `PrimitiveRingKey` payloads for every operation, and allowed a ring action to be restored without recording the exact catalog provenance.

3 Mathematical action

For normalized operation

$$g(i, \mathbf{n}) = (\pi_g(i), A_g \mathbf{n} + \tau_i^g),$$

and canonical zero-shift ring representative R_k , the derived action records

$$g(R_k) = T_{\mathbf{s}_{gk}} R_{p_g(k)},$$

plus one cycle parameterization

$$(c_{gk}, \epsilon_{gk}), \quad \epsilon_{gk} \in \{-1, +1\}.$$

`p_g(k)` is stored as an integer ring position, not as a repeated full key.

Normalized representatives compose with cocycle

$$\hat{g}\hat{h} = T_{\mathbf{c}(g,h)} \widehat{gh}.$$

Therefore exact placement composition requires

$$\mathbf{s}_{gh,k} = A_g \mathbf{s}_{h,k} + \mathbf{s}_{g,p_h(k)} - \mathbf{c}(g, h).$$

The builder verifies this identity for every operation pair and every stored ring. It also verifies composition of cycle orientation and start position.

4 Public API

```
@dataclass(frozen=True, slots=True)
class RingSymmetryImage:
    target_ring_position: int
    target_image_shift: LatticeShift
    parameterization: CycleParameterization

@dataclass(frozen=True, slots=True)
class PrimitiveRingSymmetryIndex:
    periodic_net_symmetry_digest: str
    periodic_net_view_digest: str
    topology_graph_digest: str
    primitive_ring_catalog_digest: str
```

```

complete_for_ring_sizes_up_to: int
source_search_completed_without_resource_truncation: bool
ring_keys: tuple[PrimitiveRingKey, ...]
action_table: tuple[tuple[RingSymmetryImage, ...], ...]
ring_orbits: tuple[tuple[int, ...], ...]
ring_stabilizers: tuple[tuple[int, ...], ...]
digest: str

```

Builder:

```

build_primitive_ring_symmetry_index(
    view: PeriodicNetView,
    symmetry: PeriodicNetSymmetry,
    ring_index: PrimitiveRingIndex,
    *,
    max_composition_checks: int = 5_000_000,
) -> PrimitiveRingSymmetryIndex

```

Convenience methods include:

```

ring_position(ring_key)
ring_key(position)
ring_image(operation_index, ring_key)
target_ring_key(image)
map_placement(symmetry, operation_index, placement)

```

5 Source invariants

Construction requires:

1. `symmetry.periodic_net_view_digest == view.digest`;
2. matching topology graph digests;
3. the same framework edge-orbit set in the view and ring catalog;
4. one exact action row per symmetry operation;
5. one exact image per catalog ring; and
6. a complete group-action homomorphism under the stored translation cocycle.

The index stores the catalog completeness bound but does not reinterpret it. Ring orbits are complete only for the supplied catalog.

6 Compact persistence

Schema:

```
mdstats.primitive-ring-symmetry.v1
```

Complete ring keys appear once. Table cells carry only:

```

target ring position
three-integer image shift
start position
orientation

```

For the Na-LTA gate, this reduces the ring-action JSON from repeated full-key records to approximately 1 MB while preserving all 96-by-82 exact images.

7 Serialization

`from_dict()` requires all three source objects:

```
PrimitiveRingSymmetryIndex.from_dict(
    payload,
    view=view,
    symmetry=symmetry,
    ring_index=ring_index,
)
```

The method rebuilds and revalidates the complete action table. A payload cannot be transferred to another symmetry group or another primitive-ring catalog even when the topology graph is the same.

8 Edge cases

- empty ring catalog: supported as an empty finite domain only if the action table shape is consistent;
- ring catalog with different edge set: rejected;
- different view policy on the same topology: rejected by view digest;
- catalog with same keys but different provenance: rejected by catalog digest;
- nonsymmorphic operation: image shifts are composed with the exact cocycle;
- translated input placement: mapped by $A_g \mathbf{t} + \mathbf{s}_{gk}$;
- resource limit exceeded during homomorphism validation: no partial index is returned.

9 Focused tests

- identity, rotation, and reflection actions;
- ring orbit and stabilizer partitions;
- parallel-edge and occurrence orientation handling;
- source-bound serialization;
- wrong-group and wrong-catalog rejection;
- exact translated-placement action; and
- all 82 Na-LTA rings under the 96-operation group.

10 References

- [1] S. J. Chung, Th. Hahn, and W. E. Klee, *Acta Cryst. A* **40**, 42-50 (1984), doi:10.1107/S0108767384000088.
- [2] O. Delgado-Friedrichs and M. O’Keeffe, *Acta Cryst. A* **59**, 351-360 (2003), doi:10.1107/S0108767303012017.